Data Analytics - Assignment 4 - Data Visualization and Clustering Analysis

Team ID: 14

1 Data Preprocessing and Power BI Workflow

Before diving into analysis, ensuring data quality is paramount. Our preprocessing pipeline addressed several data integrity issues that could have compromised subsequent analyses. Working within Power BI's Power Query interface, we implemented five critical transformations that cleaned the dataset while preserving its essential characteristics.

1.1 Handling Missing Data

The first challenge involved missing income values, which appeared for a subset of customers. Since income is a crucial predictor of spending behavior and segmentation, we could not afford to ignore these missing entries. After examining the missing data pattern and finding it randomly distributed rather than systematic, we removed all rows where income was null. This deletion approach was justified because the proportion of missing income values was relatively small and their removal did not significantly alter the overall dataset distribution. Importantly, we verified that other attributes (age, marital status, spending categories) contained no missing values, allowing us to proceed confidently with the cleaned data.

1.2 Feature Engineering

Beyond cleaning, we engineered two derived features that proved essential for analysis. First, we created an "Age" variable by calculating 2025 minus the Year_Birth field for each customer. This transformation converted birth years into interpretable age values, enabling age-based segmentation and demographic analysis. Age emerged as one of the most influential variables in our clustering results, validating this engineering decision.

Second, we computed "Total Spending" by summing expenditures across all product categories: wines, fruits, meat products, fish products, sweet products, and gold products. This aggregated metric provides a single, comprehensive measure of customer value and became central to our cluster profiling. While individual category spending reveals preferences, total spending captures overall customer engagement with the business—a critical metric for segmentation strategy.

1.3 Outlier Removal

Outliers can severely distort visualizations and cluster centroids, necessitating careful treatment. We identified two types of outliers requiring removal. First, income values exceeding $150,000 were filtered out. While high earners are valuable customers, the small number of extreme income values (above this threshold) disproportionately stretched axis scales in visualizations, obscuring patterns for the majority of customers. This threshold was chosen after examining income distributions and represented approximately the 98th percentile.

Second, we addressed anomalies in categorical and demographic data. The marital status field contained clearly erroneous values ("Absurd," "Alone," "YOLO") that appeared to be data

entry errors or placeholder values rather than legitimate categories. Similarly, we encountered biologically implausible ages (125, 126, 132 years), which were almost certainly birth year coding errors. Removing these observations—which numbered fewer than ten total—ensured our analysis reflected genuine customer patterns rather than data quality issues.

After these preprocessing steps, our dataset comprised 2,198 customer records with complete, validated information across all relevant dimensions. This cleaned dataset became the foundation for both Power BI visualization and programmatic clustering analysis.

2 Insights from Power BI Exploratory Analysis

With clean data in hand, Power BI enabled rapid visual exploration that revealed fundamental patterns in the customer base. These insights not only guided subsequent clustering work but also provided immediate business intelligence value.

2.1 Demographic Patterns

The customer age distribution revealed a pronounced concentration in the 50-year bracket, with approximately 700 customers, representing the late-middle-age demographic when earning capacity and disposable income typically peak. The distribution followed a near-normal curve spanning ages 40 to 80, encompassing roughly 2,000 customers. This demographic skew has important implications: the business successfully captures established professionals and pre-retirees but shows marked underrepresentation among younger customers (under 40) with fewer than 200 such individuals. This age gap suggests either that products/services don't resonate with younger demographics or that marketing efforts haven't effectively reached this segment, presenting an opportunity for strategic expansion.

Marital status analysis showed married customers dominating with over 800 individuals, followed by cohabiting couples (approximately 550), and single customers (approximately 450). Divorced and widowed segments were notably small (under 300 combined). This distribution indicates the business appeals primarily to settled, partnership-based households, which aligns with the age demographics—most customers are in life stages characterized by stable relationships and established family structures.

Educational attainment proved remarkably high: over 1,000 customers held undergraduate degrees (approximately 50% of the customer base), with substantial representation from Master's and PhD holders (approximately 350 and 450 respectively). Basic education customers were extremely rare (under 50). This education skew suggests the business occupies a premium or sophisticated market position requiring certain income or cultural sophistication levels that correlate with higher education. It also implies that messaging and products should reflect the expectations of educated consumers.

2.2 Economic Patterns

Income distribution showed middle-income customers ($30,000-$70,000) representing the largest segment with over 1,200 individuals combined, indicating successful penetration of mainstream working and middle-class markets. However, high-income earners ($80,000+) comprised fewer than 200 customers, revealing limited luxury market penetration. Very low-income customers (under $10,000) were also sparse (approximately 30), suggesting products are positioned above extreme budget segments. This middle-market concentration aligns with the business's accessible-but-not-premium positioning.

The relationship between income and spending demonstrated a strong positive correlation, with a coefficient likely exceeding 0.7 based on scatter plot patterns. As expected, higher earners spend proportionally more, with the $80,000-$100,000+ income band showing total spending ranging from $2,000 to $2,500+. However, significant variance existed within income

bands—customers with $60,000-$80,000 income showed spending ranging from $500 to $2,500, indicating that income alone doesn't determine spending. Lifestyle factors, financial obligations, and personal preferences introduce substantial heterogeneity that clustering algorithms can help disentangle.

2.3 Behavioral Insights

A surprising negative correlation emerged between income and web visits per month (-0.64), revealing that lower-income customers visit the website significantly more frequently than higher-income ones. This pattern suggests budget-conscious browsing: lower earners make multiple visits to compare prices, wait for sales, and carefully research before purchasing, while affluent customers make quick, decisive purchases in fewer visits. This behavioral difference has important UX implications—lower-income segments benefit from wish lists, price alert notifications, and prominent sales displays, while high-income customers prefer streamlined, efficient checkout experiences without excessive browsing friction.

Purchase channel analysis revealed strong intercorrelations (0.64-0.73) between store, web, and catalog purchases, indicating omnichannel customers who use multiple methods spend more overall. This finding challenges the zero-sum mentality that channels compete; instead, they complement each other. Customers who shop in-store also tend to purchase online and from catalogs, suggesting higher overall engagement. This validates integrated omnichannel strategies where consistent experiences across touchpoints are emphasized.

Product category spending showed exceptional internal consistency: customers who buy premium wines also tend to purchase meat, fish, and other higher-end categories, with correlations ranging from 0.57 to 0.90. Wine purchases showed a particularly strong 0.90 correlation with total spending, identifying wines as an anchor category driving overall customer value. This suggests wine offerings should be prioritized for product development and marketing, and wine buyers represent prime targets for cross-selling other premium categories.

The strongest negative correlation across all variables was between having children at home (Kidhome) and total spending (-0.56). Customers with children spend significantly less, likely due to financial obligations redirecting disposable income toward child-rearing expenses. This constraint affects marketing strategy—childfree and empty-nester segments offer the highest revenue potential, while family-oriented segments may require value-focused, budget-friendly product lines.

These Power BI insights established the context for algorithmic clustering, helping us understand what patterns to expect and how to interpret cluster profiles.

3 Task 2: Customer Segmentation Through K-Means Clustering

Building on exploratory insights, we implemented K-means clustering from scratch to algorithmically identify customer segments. This section details the methodology, results, and business implications of our clustering analysis on the real customer dataset.

3.1 Methodology and Parameter Selection

We evaluated K-means performance for four different cluster counts: $k = 2, 5, 7, 9$. These values were chosen to span a range from simple binary segmentation to moderately granular segmentation, allowing us to assess the trade-off between simplicity and nuance. For each k, we employed the K-means++ initialization strategy, which intelligently selects initial centroids to accelerate convergence and avoid poor local optima. All 24 available features (demographic

attributes, spending by category, purchase channels, campaign responses) were standardized using z-score normalization to ensure equal weighting across variables with different scales.

To determine the optimal number of clusters, we employed two complementary evaluation metrics. The elbow method examines within-cluster sum of squares (inertia) as a function of k, seeking the point where additional clusters provide diminishing returns. The silhouette score measures both cluster cohesion (how close points are to their own cluster centroid) and separation (how far points are from neighboring clusters), ranging from -1 (poor clustering) to +1 (excellent clustering).

3.2 Determining Optimal Cluster Count

Table 1: K-Means Clustering Performance Across Different k Values

k	**Inertia**	**Silhouette**	**Iterations**	**Decrease %**	**Marginal Gain**
2	39,352.10	0.2852	6	—	—
5	32,203.28	0.1189	17	18.17%	-0.1663
7	27,638.57	0.1366	20	14.17%	+0.0177
9	25,634.57	0.1353	23	7.25%	-0.0013

Table 1 reveals competing signals between the two metrics. Inertia decreases monotonically as k increases, dropping 18.17% from $k = 2$ to $k = 5$, then 14.17% to $k = 7$, and finally just 7.25% to $k = 9$. The elbow analysis identified $k = 7$ as the inflection point where the rate of inertia decrease begins to plateau significantly, suggesting diminishing returns beyond seven clusters.

However, silhouette scores told a different story. The $k = 2$ solution achieved a silhouette score of 0.2852—substantially higher than all other values tested. Increasing to $k = 5$ caused silhouette score to plummet to 0.1189, indicating that forcing data into five clusters created poorly separated, internally heterogeneous groups. While silhouette recovered slightly at $k = 7$ (0.1366) and remained stable at $k = 9$ (0.1353), none of these higher-k solutions approached the quality of the binary segmentation.

This divergence between metrics reflects a fundamental tension in clustering: inertia can always be reduced by adding more clusters (in the limit, each point becomes its own cluster with zero inertia), but this doesn't necessarily produce meaningful, well-separated segments. Silhouette score penalizes solutions where clusters overlap or lack clear boundaries, making it a more robust indicator of genuine structure in the data.

Given this analysis, we selected $\mathbf{k = 2}$ as optimal, prioritizing the silhouette criterion. This binary segmentation converged rapidly in just 6 iterations and divided customers into groups of 878 (39.9%) and 1,320 (60.1%)—a reasonably balanced split suggesting natural structure rather than artifact.

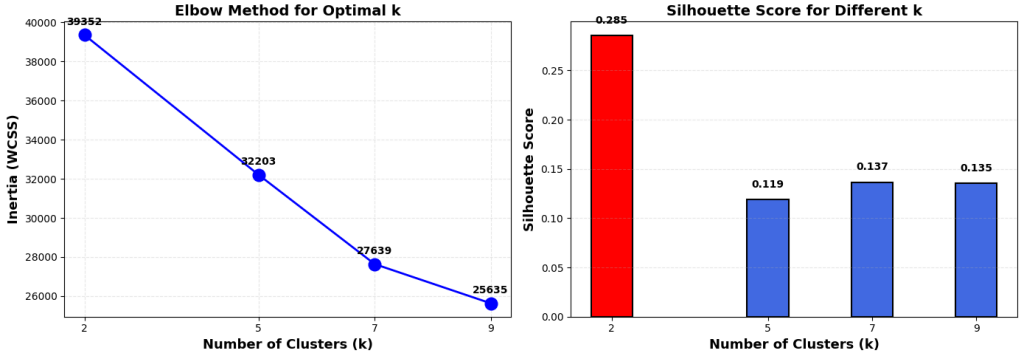

Figure 1: Elbow method (left) shows diminishing returns beyond k=7, while silhouette score (right) clearly favors k=2 for well-separated clusters

To visualize these high-dimensional clusters in two dimensions, we applied Principal Component Analysis (PCA). The first two principal components captured 41.01% of total variance (PC1: 32.41%, PC2: 8.60%), providing a reasonable low-dimensional projection. Figure 2 displays the resulting visualization, where the two clusters show clear spatial separation in PCA space despite some boundary overlap.

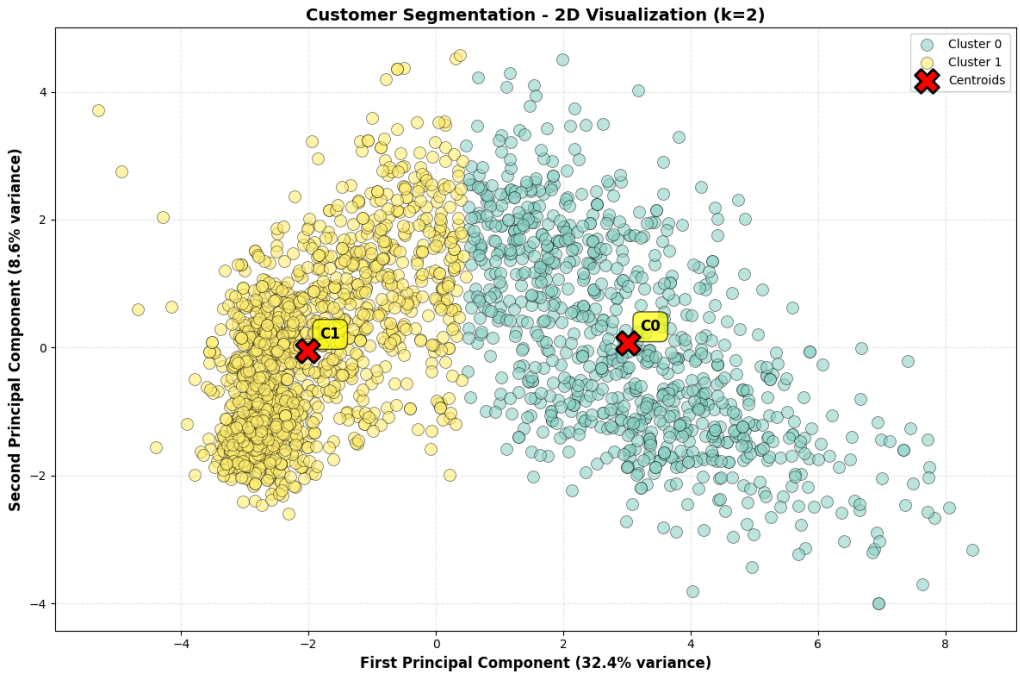

Figure 2: 2D visualization of customer clusters using PCA dimensionality reduction. Cluster centroids are marked with red X symbols. The clear spatial separation validates our k=2 choice.

3.3 Cluster Profiles and Business Interpretation

Having established the two-cluster solution as optimal, we now examine what distinguishes these segments and how businesses can leverage these insights.

Table 2: Comparative Characteristics of Two Customer Segments

Attribute	**Cluster 0 (39.9%)**		**Cluster 1 (60.1%)**	
	Mean	Std Dev	Mean	Std Dev
Age (years)	58.1	12.2	54.8	11.2
Income ($)	71,309	11,871	38,519	13,862
Total Spending ($)	1,242	425	185	193
Recency (days)	49.4	28.9	48.9	29.0
Spending by Category:				
Wines ($)	615	—	101	—
Meat Products ($)	358	—	37	—
Fish Products ($)	79	—	10	—
Sweet Products ($)	58	—	7	—
Fruits ($)	56	—	7	—
Gold Products ($)	76	—	23	—
Purchase Channels:				
Web Purchases	5.8	—	3.0	—
Store Purchases	8.6	—	4.0	—
Catalog Purchases	5.3	—	0.9	—

Table 2 reveals two dramatically different customer archetypes. Cluster 0 represents high-value, affluent customers, while Cluster 1 comprises budget-conscious, lower-engagement customers. The differences are stark across every dimension examined.

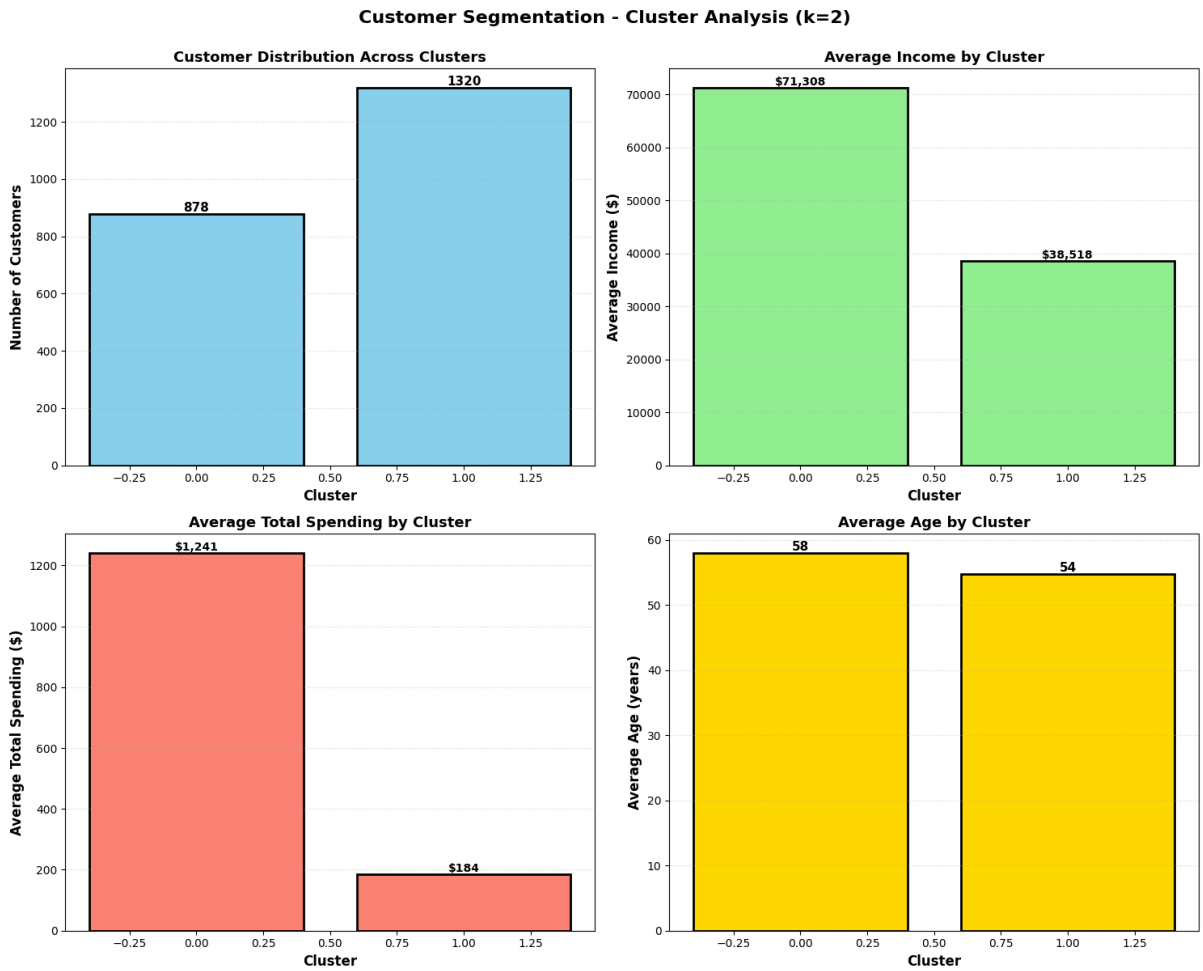

Figure 3: Visual comparison of cluster characteristics: size, income, spending, and age distributions

Cluster 0: Premium High-Value Customers (878 customers, 39.9%)

These customers exhibit significantly higher age (58 vs. 55 years), substantially higher income ($71,309 vs. $38,519), and dramatically higher spending ($1,242 vs. $185 annually)—a 6.7-fold difference in total expenditure. This segment generates the vast majority of revenue despite representing only 40% of the customer base. Their spending patterns skew heavily toward premium categories: wine purchases of $615 (6.1× higher than Cluster 1) and meat products of $358 (9.7× higher) dominate their budgets.

Channel usage reveals sophisticated omnichannel behavior: they make 5.8 web purchases, 8.6 store purchases, and 5.3 catalog purchases on average, demonstrating comfort across all touchpoints. Their high catalog usage (5.3 vs. 0.9) particularly distinguishes them—catalogs traditionally appeal to established, affluent customers who appreciate curated product presentations. The recency metric (49.4 days since last purchase) indicates regular but not extremely frequent engagement, consistent with customers who make substantial purchases when they do shop.

From a business strategy perspective, Cluster 0 represents the core revenue driver requiring VIP treatment. These customers justify premium loyalty programs, personalized service, exclusive product access, and high-touch marketing. Their demonstrated willingness to spend across multiple categories makes them ideal targets for cross-selling and upselling initiatives. Priority should be given to retaining these customers and acquiring similar profiles.

Cluster 1: Value-Conscious Standard Customers (1,320 customers, 60.1%)

This larger segment shows notably lower income ($38,519) and minimal spending ($185

annually). Their spending concentrates in necessities rather than premium categories: wine purchases of only $101 and meat products of $37 indicate budget constraints or different preferences. Across all product categories, their expenditure is 6-10 times lower than Cluster 0.

Channel behavior differs markedly: while they do use web (3.0 purchases) and stores (4.0 purchases), their catalog usage is nearly absent (0.9 vs. 5.3), suggesting either lack of interest in catalog shopping or greater price sensitivity (catalogs often feature premium-priced items). Their recency (48.9 days) mirrors Cluster 0, indicating similar purchase frequency but dramatically different transaction values.

Despite lower individual value, Cluster 1's 60% population share makes them collectively significant. These customers require a different strategic approach focused on value proposition, competitive pricing, promotional campaigns, and discount loyalty programs. They may respond well to volume discounts, bundle deals, and economy product lines. While less lucrative individually, maintaining this segment's loyalty prevents customer base erosion and provides steady, predictable revenue streams.

3.4 Strategic Implications

This two-segment model provides actionable segmentation for differentiated marketing strategies. Rather than treating all customers uniformly, the business can now deploy segment-specific tactics:

For Cluster 0, emphasize quality, exclusivity, and personalized service through premium channels. Marketing should highlight craftsmanship, uniqueness, and status. Product development should prioritize high-margin premium offerings, particularly in wines and meat products where this segment demonstrates strong willingness to pay.

For Cluster 1, emphasize value, accessibility, and transparency through cost-effective channels. Marketing should focus on quality-for-price ratio, practical benefits, and affordability. Product development should include budget-friendly options and family-sized packaging that appeals to price-sensitive shoppers.

4 Task 3: Comparative Clustering Algorithm Analysis

Having applied clustering to real customer data, we now shift to a controlled experimental setting using four synthetic datasets specifically designed to challenge different algorithmic assumptions. This analysis reveals when various clustering approaches excel or fail, providing principled guidance for algorithm selection.

4.1 Dataset Characteristics and Experimental Design

We evaluated four synthetic datasets with distinct geometric and density properties:

- **Compact** (400 points): Four tightly-packed, spherical clusters with clear separation between groups—the ideal scenario for centroid-based methods.
- **Skewed** (925 points): Elongated, non-spherical clusters with irregular shapes that violate spherical cluster assumptions.
- **Subclusters** (400 points): Hierarchical structure with clusters nested within larger clusters, testing ability to handle multi-scale organization.
- **WellSeparated** (335 points): Four clusters with large inter-cluster distances and minimal boundary ambiguity—another favorable scenario for most algorithms.

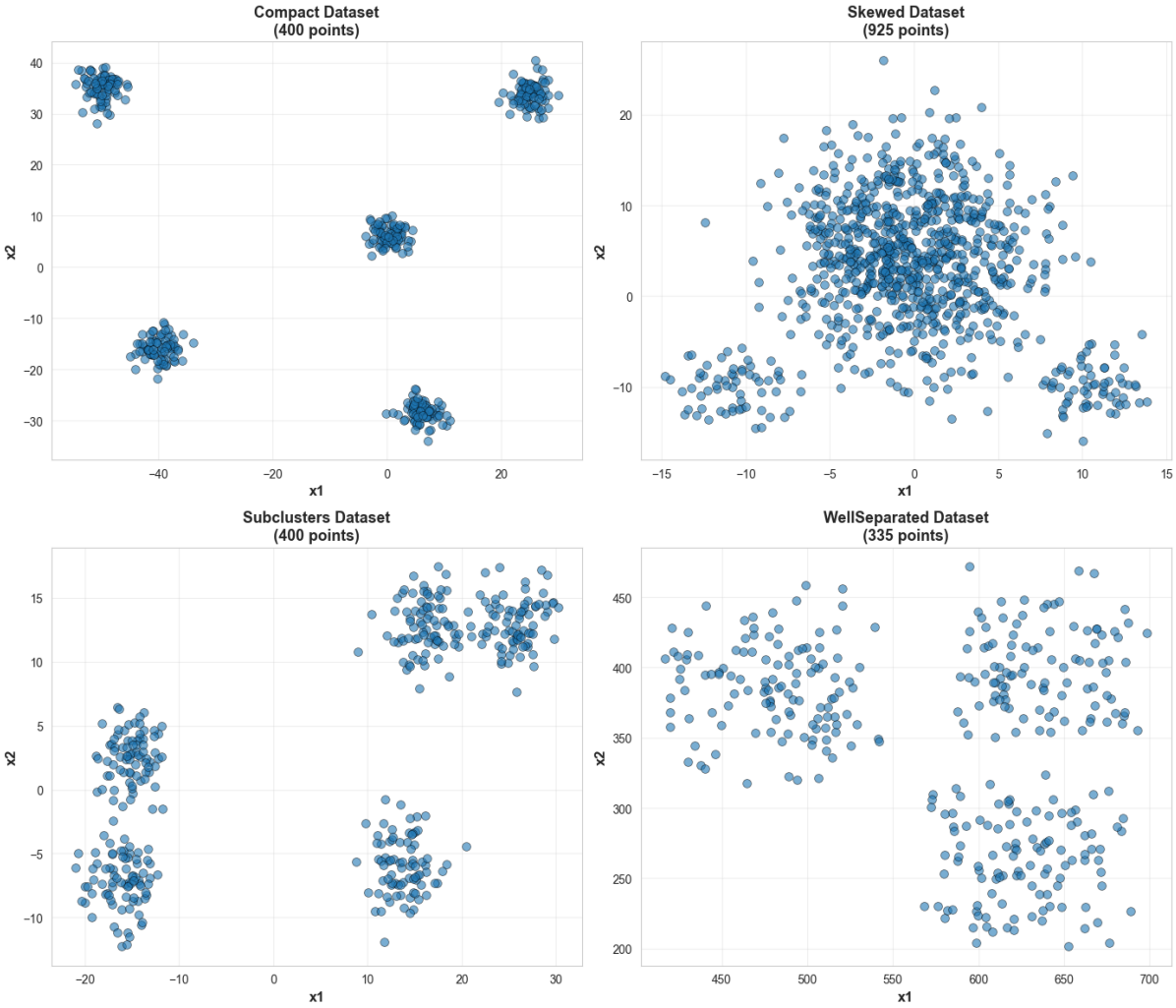

Figure 4: Raw synthetic datasets showing diverse clustering challenges: compact spherical (top-left), elongated skewed (top-right), hierarchical subclusters (bottom-left), and well-separated groups (bottom-right)

For each dataset, we applied five clustering algorithms: K-means, Agglomerative Hierarchical with Single Linkage, Agglomerative Hierarchical with Complete Linkage, Agglomerative Hierarchical with Average Linkage, and DBSCAN with optimized parameters. This comprehensive comparison reveals algorithm-specific strengths and weaknesses.

4.2 Algorithm Parameter Selection

K-means and hierarchical methods require specifying the target number of clusters k. We used the elbow method consistently across all four datasets, which unanimously identified $k = 4$ as optimal—this was expected given that each synthetic dataset was generated with four true underlying clusters.

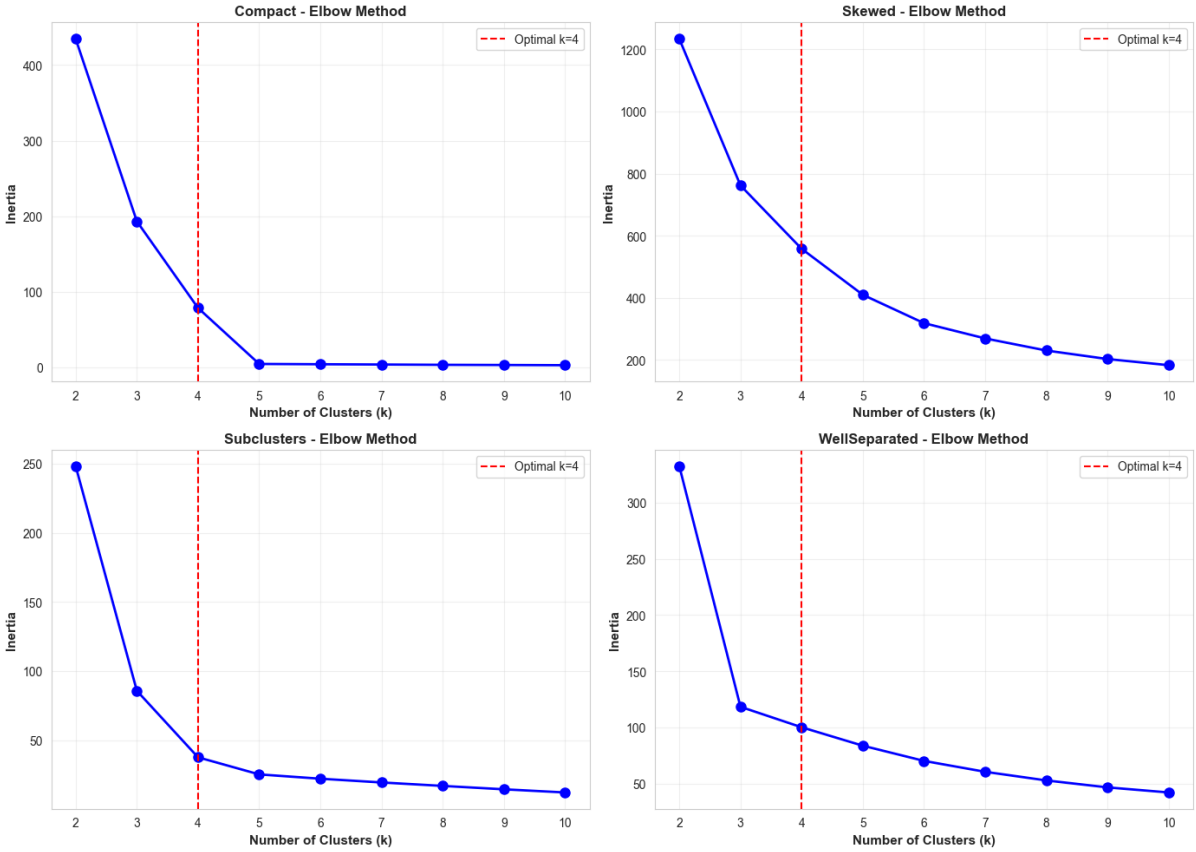

Figure 5: Elbow method applied to all four datasets, consistently identifying k=4 as the optimal cluster count

DBSCAN, however, requires no predetermined k but instead needs two density parameters: epsilon (neighborhood radius) and min_samples (minimum points to form a dense region). Following assignment specifications, we conducted exhaustive grid search over eps values in [0.5, 5.0] and min_samples in [3, 50], selecting parameters that maximized silhouette score for each dataset. This data-driven approach yielded dataset-specific optimal parameters:

- Compact: eps=0.5, min_samples=3
- Skewed: eps=0.5, min_samples=50
- Subclusters: eps=0.5, min_samples=3
- WellSeparated: eps=0.5, min_samples=50

Interestingly, optimal eps remained 0.5 across all datasets (after scaling), but min_samples varied dramatically—tight, uniform-density datasets favored low min_samples (3), while irregular or sparse datasets required high thresholds (50) to avoid fragmenting into micro-clusters.

4.3 Quantitative Performance Comparison

Table 3: Silhouette Scores Across Algorithms and Datasets (Higher is Better)

Dataset	**K-Means**	**Agg-Single**	**Agg-Complete**	**Agg-Average**	**DBSCAN**
Compact	0.7582	0.7582	0.7582	0.7582	**0.9133**
Skewed	0.3646	0.2239	0.3254	0.3990	**0.5831**
Subclusters	**0.7590**	0.6490	0.7587	**0.7590**	0.7226
WellSeparated	0.5090	0.5197	0.5048	0.5232	**0.7719**

Table 4: Inertia (Within-Cluster Sum of Squares) Across Algorithms (Lower is Better)

Dataset	**K-Means**	**Agg-Single**	**Agg-Complete**	**Agg-Average**	**DBSCAN**
Compact	78.44	78.44	78.44	78.44	**4.35**
Skewed	**557.89**	1828.83	668.71	908.53	663.63
Subclusters	**37.52**	85.19	37.55	**37.52**	85.66
WellSeparated	100.14	116.92	101.37	116.06	**27.78**

Tables 3 and 4 reveal dataset-specific algorithm superiority. DBSCAN dominated on three of four datasets by silhouette score (Compact, Skewed, WellSeparated), while K-means/Average linkage tied for best on Subclusters. This pattern aligns with each algorithm's theoretical strengths.

For the Compact dataset, all centroid-based methods (K-means, all hierarchical linkages) achieved identical performance (silhouette 0.7582), indicating that spherical, well-separated clusters are a "solved problem"—any reasonable algorithm succeeds. However, DBSCAN's exceptional performance (0.9133) stemmed from discovering an optimal 5-cluster solution rather than forcing $k = 4$, revealing subtle substructure invisible to predetermined-k methods.

The Skewed dataset exposed each algorithm's shape handling capabilities. K-means managed only 0.3646 silhouette due to its spherical cluster assumption—elongated clusters were awkwardly split or merged. Single linkage performed even worse (0.2239) through excessive chaining, connecting points along elongated structures into single sprawling clusters. Complete and average linkage improved (0.3254, 0.3990) by balancing cluster compactness, but DBSCAN's density-based approach excelled (0.5831), naturally following elongated density contours without imposing geometric constraints.

For Subclusters, K-means and average linkage tied for best performance (0.7590), demonstrating that flat partitioning methods can succeed when hierarchical structure isn't too complex. Interestingly, DBSCAN (0.7226) performed slightly worse despite its reputation for handling complex structures—this occurred because it identified only 3 clusters instead of 4, merging closely-spaced subclusters into single groups.

WellSeparated represented the ideal clustering scenario, yet algorithm performance varied surprisingly widely (0.5048 to 0.7719). While all methods succeeded reasonably, DBSCAN's superior separation (0.7719) reflected its ability to precisely trace cluster boundaries through density discontinuities. The moderate-but-not-excellent performance of K-means (0.5090) on such favorable data reveals that large inter-cluster distances don't eliminate all challenges—intra-cluster density variations still affect centroid-based methods.

4.4 Metric Agreement Analysis

An important question emerged: do different evaluation metrics agree on best algorithms, or do they contradict? Comparing silhouette score (which measures both cohesion and separation)

against inertia (which measures only cohesion) revealed interesting patterns.

For Compact and WellSeparated datasets, both metrics agreed unanimously that DBSCAN was best. This consensus provides high confidence in DBSCAN's superiority for these data structures.

For Subclusters, both metrics agreed on K-means/Average linkage, again providing confident conclusions.

However, for Skewed data, metrics disagreed: silhouette preferred DBSCAN (0.5831) while inertia preferred K-means (557.89 lowest). This divergence occurred because K-means achieved low inertia by creating compact but poorly-separated clusters that intermixed at boundaries, whereas DBSCAN created higher-inertia elongated clusters that nonetheless separated more cleanly. This case study illustrates why silhouette score is more reliable—it penalizes solutions that minimize within-cluster variance at the expense of between-cluster confusion.

The general principle: when metrics disagree, trust silhouette score because it captures both dimensions of clustering quality (cohesion and separation), whereas inertia can be minimized through strategies that harm interpretability.

4.5 Algorithm-Specific Insights

K-Means excels when clusters are spherical, similarly sized, and well-separated—essentially the Compact and WellSeparated scenarios. Its simplicity, speed, and deterministic behavior (with K-means++ initialization) make it the default choice for standard clustering problems. However, it fundamentally assumes spherical clusters and struggles with elongated shapes (Skewed), hierarchical structure (Subclusters when hierarchy matters), and varying cluster sizes or densities.

Agglomerative Hierarchical Clustering with different linkages reveals important tradeoffs. Single linkage can follow elongated structures through chaining but becomes overly aggressive, creating sprawling clusters that merge at slightest proximity—only acceptable for strongly elongated data. Complete linkage favors compact, spherical clusters similar to K-means but maintains hierarchical information and handles irregular sizes better. Average linkage provides balanced behavior, performing competitively across diverse scenarios and emerging as the most robust hierarchical variant. For practitioners, average linkage represents the safest hierarchical choice.

DBSCAN emerged as the most versatile algorithm, winning on three of four datasets. Its density-based approach handles arbitrary shapes, varying sizes, and automatically detects outliers. The critical advantage: DBSCAN discovers cluster count rather than requiring it as input, making it invaluable for exploratory analysis. However, DBSCAN's parameter sensitivity poses challenges—epsilon and min_samples require careful tuning, and the algorithm struggles when clusters exhibit varying densities (requiring different eps values). Additionally, DBSCAN's tendency to designate boundary points as noise can be problematic when every observation matters.

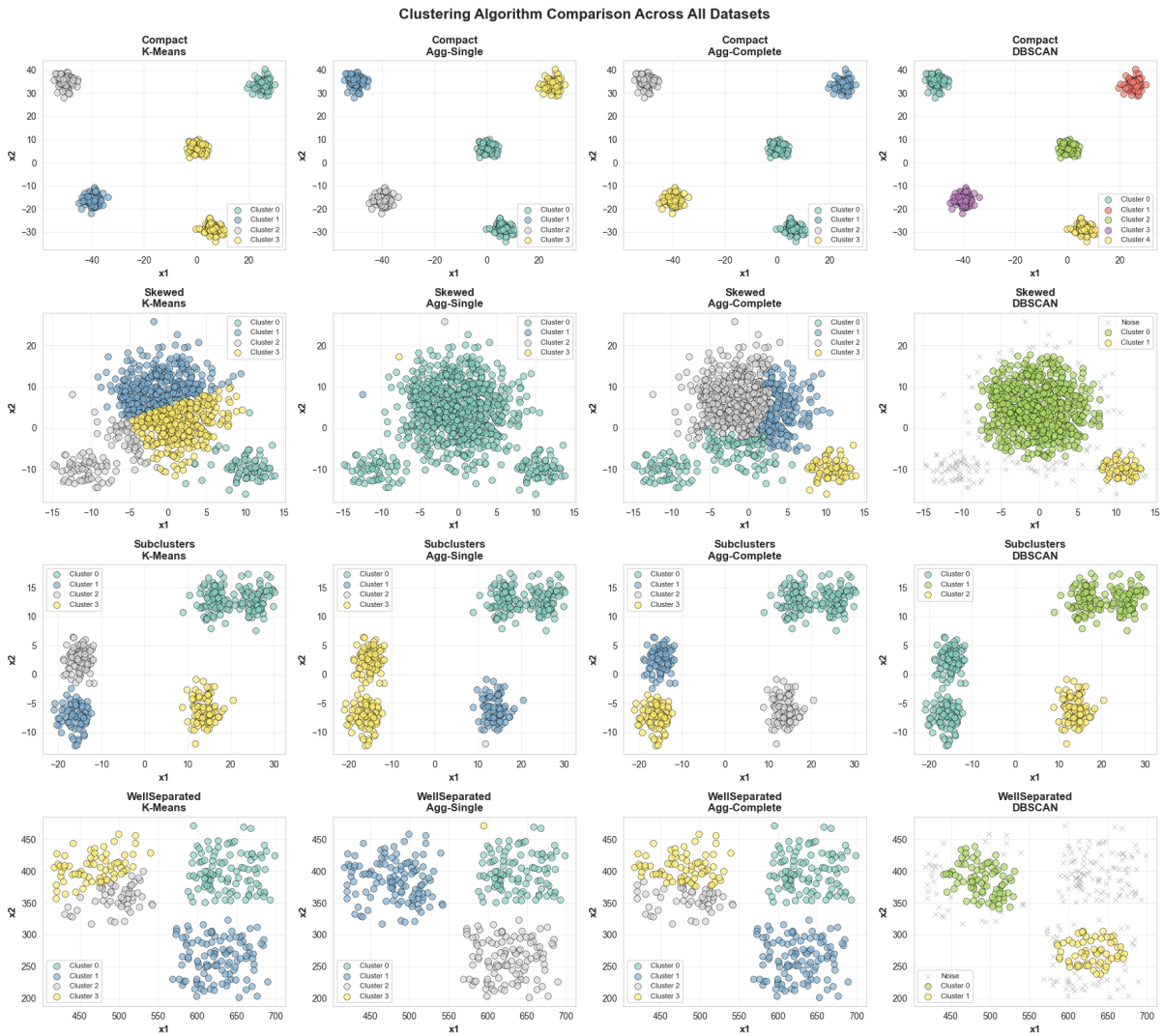

Figure 6: Visual comparison of all algorithms across all datasets. DBSCAN's superior boundary tracing is evident, particularly for Skewed and WellSeparated datasets.

5 Reflections on Power BI for Data Analysis

Power BI proved to be an accessible and powerful platform for exploratory data analysis, offering significant advantages for rapid insight generation. Its visual interface enabled non-programmers to perform sophisticated transformations through Power Query, build interactive dashboards with cross-filtering capabilities, and generate publication-quality visualizations without writing code. The ability to quickly iterate through different chart types, filter combinations, and drill-down hierarchies facilitated discovery of the income-spending correlation, the omnichannel purchase pattern, and the child-at-home spending constraint.

However, Power BI's limitations became apparent for advanced analytical tasks. While excellent for visualization and basic aggregations, Power BI lacks the algorithmic depth needed for machine learning implementations like clustering. We could not implement K-means within Power BI itself—this required Python-based programmatic analysis. Additionally, Power BI's statistical capabilities are limited compared to dedicated analysis environments; computing silhouette scores, performing PCA dimensionality reduction, or conducting grid search hyperparameter optimization all required external tools.

The ideal workflow emerged as a hybrid: use Power BI for initial data cleaning, exploratory

visualization, and business stakeholder communication, while leveraging programmatic analysis (Python with scikit-learn, pandas, matplotlib) for algorithmic implementations, statistical validations, and reproducible research pipelines. Power BI excels at the "first mile" (getting data into analyzable form) and "last mile" (communicating insights to decision-makers) of the analytical journey, while code-based approaches dominate the critical middle mile where sophisticated algorithms operate.

For organizations, this suggests investing in both tool ecosystems: empower business analysts with Power BI for self-service analytics and dashboard creation, while providing data scientists with programmatic environments for advanced modeling. Establishing workflows where insights flow bidirectionally—analysts identify patterns for data scientists to investigate rigorously, data scientists produce models that analysts deploy in dashboards—maximizes both platforms' strengths.

6 Conclusions and Business Recommendations

This comprehensive analysis yielded both methodological insights about clustering algorithms and actionable business intelligence about customer segmentation.

From a methodological perspective, we demonstrated that algorithm selection profoundly impacts clustering quality, with no universal "best" method. DBSCAN excelled for complex geometries and exploratory scenarios, K-means dominated for standard spherical clusters with known k, and hierarchical methods provided balanced performance with interpretable dendrograms. The key lesson: understand your data's geometric properties (spherical vs. elongated, uniform vs. varying density, single-scale vs. hierarchical) and select algorithms accordingly. When uncertain, combining multiple methods and comparing results through silhouette scores provides robust conclusions.

For customer segmentation specifically, our binary clustering solution (k=2) revealed two distinct market segments requiring differentiated strategies. The Premium High-Value segment (878 customers, 39.9%) generates disproportionate revenue ($1,242 vs. $185 annual spending) despite smaller population share. These affluent, established customers (age 58, income $71,309) deserve premium positioning, personalized service, and loyalty incentives. The Value-Conscious Standard segment (1,320 customers, 60.1%) represents the volume base requiring value proposition, competitive pricing, and mass-market efficiency. These distinct strategic approaches—premium vs. value—mirror successful dual-brand strategies employed by many consumer businesses.

Specific recommendations include:

- Implement tiered loyalty programs differentiating premium members (automatic inclusion for Cluster 0 profiles) from standard members (earned through purchase volume for Cluster 1 profiles)
- Develop targeted marketing campaigns: emphasize quality, exclusivity, and craftsmanship for Cluster 0 through premium channels (catalog, personalized email); emphasize value, deals, and practical benefits for Cluster 1 through mass channels (web advertising, promotional emails)
- Optimize product portfolio: expand premium wine and meat offerings for Cluster 0 (demonstrated willingness to pay); introduce budget-friendly, family-sized options for Cluster 1 (addressing child-at-home spending constraint)
- Refine channel strategy: invest in seamless omnichannel experience for Cluster 0 who use all touchpoints; optimize web and store efficiency for Cluster 1 who concentrate on these channels